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VEHICLE HEADLAMP

Abstract

A vehicle headlamp includes a light source that emits light, a reflector that reflects the light emitted from the light source, a shade that blocks a portion of the light reflected by the reflector, and a projection lens that projects the light reflected by the reflector. The shade is provided with a control end located on a side of the light source. The control end includes a reflective control surface subjected to a reflective treatment and having a focal point of light positioned thereon or in the vicinity thereof, and a sloped surface subjected to a reflective treatment and continuous with an edge of the reflective control surface on the side of the light source. The sloped surface is inclined with respect to the reflective control surface to be displaced downward as a distance from the reflective control surface increases.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based on and claims priority from Japanese Patent Application No. 2024-025174, filed on Feb. 22, 2024, with the Japan Patent Office, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a technical field concerning a vehicle headlamp in which light emitted from a light source is reflected by a reflector and a portion of the light is blocked by a shade.

BACKGROUND

[0003] There is a type of vehicle headlamp in which a lamp unit, which includes a reflector and a shade, is arranged in the interior of a lighting device outer case, which is constituted with a cover and a lamp housing (see, e.g., Japanese Patent Laid-Open Publication No. 2012-059643). In the vehicle headlamp described in Japanese Patent Laid-Open Publication No. 2012-059643, a reflector, a shade, a projection lens, and other components are arranged in the interior of the lighting device outer case, light emitted from a light source is reflected by the reflector toward the shade where a portion of the light is blocked, while the unblocked light is controlled and directed outward as substantially parallel rays from the projection lens.

SUMMARY

[0004] In the vehicle headlamp equipped with the shade as described in Japanese Patent Laid-Open Publication No. 2012-059643, since the focal point of light reflected by the reflector is positioned on or near the surface of one end (control end) of the shade on the light source side, heat may be accumulated in the control end depending on the incidence of light.

[0005] In particular, when the angle of incidence of light with respect to an incident surface is or is close to a perpendicular angle, the illuminance is likely to increase, which may cause issues such as melting and discoloration in the area with the increased illuminance.

[0006] Therefore, a vehicle headlamp of the present disclosure aims to prevent melting and discoloration in the control end.

[0007] According to one aspect of the present disclosure, a vehicle headlamp includes a light source that emits light, a reflector that reflects the light emitted from the light source, a shade that blocks a portion of the light reflected by the reflector, and a projection lens that projects the light reflected by the reflector. The shade is provided with a control end located on a side of the light source. The control end includes a reflective control surface subjected to a reflective treatment and having a focal point of light positioned thereon or in the vicinity thereof, and a sloped surface subjected to a reflective treatment and continuous with an edge of the reflective control surface on the side of the light source. The sloped surface is inclined with respect to the reflective control surface to be displaced downward as a distance from the reflective control surface increases.

[0008] Thus, the angle of incidence of light with respect to the sloped surface, which is inclined with respect to the reflective control surface, becomes smaller.

[0009] According to another aspect of the present disclosure, a vehicle headlamp includes a light source that emits light, a reflector that reflects the light emitted from the light source, a shade that blocks a portion of the light reflected by the reflector, and a projection lens that projects the light reflected by the reflector. The shade is provided with a control end located on a side of the light source. The control end includes a reflective control surface subjected to a reflective treatment and having a focal point of light positioned thereon or in the vicinity thereof, and a tip surface that enables incidence of light in a direction different from the reflective control surface. The shade is made of a transparent material.

[0010] Thus, the incident light on the tip surface passes through the control end.

[0011] According to the present disclosure, it is possible to prevent melting and discoloration in the control end.

[0012] The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 illustrates, along with FIGS. 2 to 4, a first embodiment of a vehicle headlamp according to the present disclosure, in a cross-sectional view of the vehicle headlamp.

[0014] FIG. 2 is a cross-sectional view of a shade.

[0015] FIG. 3 is a cross-sectional view illustrating an example in which the shade is made of a transparent material.

[0016] FIG. 4 is a cross-sectional view of the shade illustrating another example of a control end.

[0017] FIG. 5 illustrates, along with FIG. 6, a second embodiment of a vehicle headlamp according to the present disclosure, in a cross-sectional view of the vehicle headlamp.

[0018] FIG. 6 is a cross-sectional view of a shade.

DETAILED DESCRIPTION

[0019] In the following detailed description, reference is made to the accompanying drawings which form a part hereof. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made without departing from the spirit or scope of the subject matter presented herein.

[0020] Hereinafter, embodiments for implementing a vehicle headlamp of the present disclosure will be described with reference to the accompanying drawings.

[0021] First, a vehicle headlamp 1 according to a first embodiment will be described (see, e.g., FIGS. 1 to 4).

[0022] The vehicle headlamp 1 includes a lamp housing 2 with an opening at a front end thereof and a cover 3 that closes the opening of the lamp housing 2 (see, e.g., FIG. 1). The lamp housing 2 and the cover 3 constitute a lighting device outer case 4, and the internal space of the lighting device outer case 4 is formed as a lamp chamber 5.

[0023] A lamp unit 6 is arranged in the lamp chamber 5. The lamp unit 6 includes a heat sink 7, a light source 8, a reflector 9, a shade 10, and a projection lens 11.

[0024] The heat sink 7 is made of, for example, a metallic material with high thermal conductivity, and includes a plate-shaped base portion 7a that faces upward and downward and a plurality of heat dissipation fins 7b protruding downward from the base portion 7a. The heat dissipation fins 7b are arranged, for example, side by side in the left-right direction.

[0025] As for the light source 8, for example, a light-emitting diode (LED) is used. The light source 8 is mounted, for example, on a circuit board (not illustrated). The circuit board is disposed on the base portion 7a of the heat sink 7.

[0026] The reflector 9 is attached at a rear lower end thereof, for example, to a rear end of the base portion 7a. An inner surface of the reflector 9 is formed as a reflective surface 9a. Light emitted from the light source 8 is reflected by the reflective surface 9a toward the shade 10.

[0027] The shade 10 is made of, for example, a colored resin material such as polycarbonate, and includes an inclined portion 12 extending diagonally with respect to the horizontal and vertical directions, and a control end 13 continuous with a rear end of the inclined portion 12 (see, e.g.,

FIGS. 1 and 2). The inclined portion **12** is displaced upward to the rear, while the control end **13** protrudes rearward from the rear end of the inclined portion **12**.

[0028] The control end **13** includes an upward-facing reflective control surface **13a**, a sloped surface **13b** continuous with the rear edge of the reflective control surface **13a**, a tip surface **13c** continuous with the rear edge of the sloped surface **13b**, and a lower surface **13d** continuous with the lower edge of the tip surface **13c** (see, e.g., FIG. 2).

[0029] The front edge of the reflective control surface **13a** is continuous with the rear edge of an upper surface of the inclined portion **12**, and a reflective treatment **14** such as metal vapor deposition is applied to the reflective control surface **13a**. Thus, incident light on the reflective control surface **13a** is reflected. The focal point of light reflected by the reflective surface **9a** of the reflector **9** is positioned on or near the reflective control surface **13a**. Therefore, the reflective surface **9a** is a necessary surface for ensuring proper optical performance in the vehicle headlamp **1** and requires a certain length in the front-rear direction.

[0030] The sloped surface **13b** is inclined with respect to the reflective control surface **13a**, so as to be displaced downward to the rear. The reflective treatment **14** such as metal vapor deposition is applied to the sloped surface **13b**. Thus, incident light on the sloped surface **13b** is reflected.

[0031] The tip surface **13c** is formed as a rearward-facing surface. The reflective treatment is not applied to the tip surface **13c** and the lower surface **13d**.

[0032] Further, in the shade **10**, the reflective treatment **14** may also be applied to an upper surface **12a** of the inclined portion **12**. By applying the reflective treatment **14** to the upper surface **12a**, it is possible to use the upper surface **12a**, for example, as a reflective surface for generating overhead sign light or other purposes.

[0033] The projection lens **11** is positioned on the side of the shade **10** opposite the light source **8**. At least light controlled by the control end **13** of the shade **10** enters the projection lens **11**. The projection lens **11** has the function of controlling and projecting incident light into approximately parallel rays.

[0034] In the vehicle headlamp **1** configured as described above, when light is emitted from the light source **8**, the emitted light is reflected by the reflective surface **9a** of the reflector **9** toward the shade **10**, where a portion of the light is blocked. The unblocked light is controlled by the control end **13** of the shade **10**, or passes near the control end **13** to enter the projection lens **11**. The projection lens **11** controls the light into approximately parallel rays, and projects them forward through the cover **3**.

[0035] When light is emitted from the light source **8**, a portion of light reflected by the reflective surface **9a** of the reflector **9** is directed as light A in an approximately horizontal state toward the control end **13**. The light A, directed in an approximately horizontal state toward the control end **13**, enters the sloped surface **13b** and the tip surface **13c** of the control end **13**. However, because the sloped surface **13b** is inclined, the light A enters the sloped surface **13** in directions other than the perpendicular direction.

[0036] At this time, if the sloped surface **13b** were not formed and the control end **13** instead had a shape composed of a rearward-facing surface M1 extending from the tip surface **13c** and an upward-facing surface M2 extending from the reflective control surface **13a** as illustrated by the dotted lines in FIG. 2, the light A would enter the small-area surface M1 from a direction perpendicular (approximately perpendicular) thereto, which would result in a high illuminance in the surface M1.

[0037] However, because the control end **13** of the vehicle headlamp **1** includes the sloped surface **13b** instead of the surfaces M1 and M2, the light A enters the sloped surface **13b**, which has a larger area than the surface M1, in directions other than the perpendicular direction. This reduces the light intensity per unit area on the sloped surface **13b**, resulting in a reduction in illuminance. Consequently, heat accumulation in the control end **13** is minimized even when the light A enters the control end **13**.

[0038] As described above, in the vehicle headlamp **1**, the control end **13** of the shade **10** includes the reflective control surface **13a** with the reflective treatment **14** applied, on or near which the focal point of light is positioned, and the sloped surface **13b** with the reflective treatment **14** applied, which is continuous with the edge of the reflective control surface **13a** on the light source **8** side, and the sloped surface **13b** is inclined with respect to the reflective control surface **13a** so as to be displaced downward with an increasing distance from the reflective control surface **13a**.

[0039] Thus, the angle of incidence of light on the large-area sloped surface **13b** inclined with respect to the reflective control surface **13a** is reduced, lowering the illuminance on the sloped surface **13b**. This minimizes heat accumulation in the control end **13**, mitigating issues such as melting and discoloration caused by the peeling of reflective treatment **14** due to heat in the control end **13**.

[0040] Further, the control end **13** is formed with the tip surface **13c**, which is continuous with the edge of the sloped surface **13b** on the light source **8** side and faces a direction different from that of the sloped surface **13b**, for example, faces rearward.

[0041] Thus, with the formation of the tip surface **13c**, the edge of the control end **13** on the light source **8** side does not form a sharp protrusion (protrusion T illustrated in FIG. 2). This ensures high strength of the control end **13** while preventing melting and discoloration in the control end **13**.

[0042] Further, in the vehicle headlamp **1**, the shade **10** may also be made of a transparent material (see, e.g., FIG. 3). When the shade **10** is made of a transparent material, light B reaching the tip surface **13c** enters the interior of the control end **13** from the tip surface **13c**, and at least a portion of the light passes through the control end **13**. Consequently, by allowing at least a portion of the light B to pass through the control end **13**, heat accumulation in the control end **13** is minimized. Additionally, the sloped surface **13b** lowers the illuminance, further minimizing heat accumulation in the control end **13**. With the shade **10** made of a transparent material, heat accumulation in the control end **13** may be effectively minimized.

[0043] This configuration sufficiently minimizes heat accumulation in the control end **13**.

Therefore, the area of the sloped surface **13b** may be reduced, allowing the area of the reflective control surface **13a** to be increased, thereby improving the strength of the control end **13**.

Consequently, the high strength of the control end **13** is ensured while sufficiently minimizing heat accumulation in the control end **13**.

[0044] Further, in the shade **10**, the rear edge of the sloped surface **13b** may be aligned with the lower edge (rear edge of the lower surface **13d**) of the control end **13**, and the control end **13** may be configured without the tip surface **13c** (see, e.g., FIG. 4). By adopting this configuration of the shade **10**, the area of the sloped surface **13b** may be maximized, significantly reducing the illuminance of incident light on the sloped surface **13b**. This may efficiently prevent melting and discoloration of the control end **13**.

[0045] Next, a vehicle headlamp **1A** according to a second embodiment will be described (see, e.g., FIGS. 5 and 6).

[0046] Moreover, the vehicle headlamp **1A** described below differs from the aforementioned vehicle headlamp **1** only in the configuration of the shade. Therefore, only the differences from the vehicle headlamp **1** will be described in detail, while other components will be designated by the same reference numerals as those used for similar components in the vehicle headlamp **1**, and description thereof will be omitted.

[0047] A lamp unit **6A** is arranged in the lamp chamber **5**. The lamp unit **6A** includes the heat sink **7**, the light source **8**, the reflector **9**, a shade **10A**, and the projection lens **11**.

[0048] The shade **10A** is made of a transparent material and includes the inclined portion **12** and a control end **13A**. The control end **13A** protrudes rearward from the rear end of the inclined portion **12**.

[0049] The control end **13A** includes the upward-facing reflective control surface **13a**, the

rearward-facing tip surface **13c** continuous with the rear edge of the reflective control surface **13a**, and the downward-facing lower surface **13d** continuous with the lower edge of the tip surface **13c**. The reflective treatment **14** is applied to the reflective control surface **13a**, but not applied to the tip surface **13c** and the lower surface **13d**.

[0050] In the vehicle headlamp **1A** configured as described above, when light is emitted from the light source **8**, the emitted light is reflected by the reflective surface **9a** of the reflector **9** toward the shade **10A**, where a portion of the light is blocked by the shade **10A**. The unblocked light is controlled by the control end **13a** of the shade **10A**, or passes near the control end **13a** to enter the projection lens **11**. The projection lens **11** controls the light into approximately parallel rays, and projects them forward through the cover **3**.

[0051] When light is emitted from the light source **8**, a portion of light reflected by the reflective surface **9a** of the reflector **9** enters the interior of the control end **13a** through the tip surface **13c**. At least a portion of the light passes through the control end **13a**. Consequently, heat accumulation in the control end **13a** is minimized.

[0052] As described above, in the vehicle headlamp **1A**, the control end **13a** of the shade **10A** includes the reflective control surface **13a** with the reflective treatment **14** applied, on or near which the focal point of light is positioned, and the tip surface **13c** that enables light to enter in a direction different from the reflective control surface **13a**. The shade **10A** is made of a transparent material.

[0053] Consequently, light entering the tip surface **13c** passes through the control end **13a**, reducing the likelihood of heat accumulation in the control end **13a**. This may prevent melting and discoloration caused by the peeling of reflective treatment **14** due to heat in the control end **13a**.

[0054] From the foregoing, it will be appreciated that various embodiments of the present disclosure have been described herein for purposes of illustration, and that various modifications may be made without departing from the scope and spirit of the present disclosure. Accordingly, the various embodiments disclosed herein are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

Claims

1. A vehicle headlamp comprising: a light source configured to emit light; a reflector configured to reflect the light emitted from the light source; a shade configured to block a portion of the light reflected by the reflector; and a projection lens configured to project the light reflected by the reflector, wherein the shade is provided with a control end located on a side of the light source, the control end includes a reflective control surface subjected to a reflective treatment and having a focal point of light positioned thereon or in a vicinity thereof, and a sloped surface subjected to a reflective treatment and continuous with an edge of the reflective control surface on the side of the light source, and the sloped surface is inclined with respect to the reflective control surface to be displaced downward as a distance from the reflective control surface increases.
2. The vehicle headlamp according to claim 1, wherein the control end is formed with a tip surface continuous with an edge of the sloped surface on the side of the light source and oriented in a direction different from that of the sloped surface.
3. The vehicle headlamp according to claim 2, wherein the shade is made of a transparent material, and enables light to enter the control end from the tip surface.
4. The vehicle headlamp according to claim 1, wherein the edge of the sloped surface on the side of the light source is aligned with a lower edge of the control end.
5. A vehicle headlamp comprising: a light source configured to emit light; a reflector configured to reflect the light emitted from the light source; a shade configured to block a portion of the light reflected by the reflector; and a projection lens configured to project the light reflected by the reflector, wherein the shade is provided with a control end located on a side of the light source, the

control end includes a reflective control surface subjected to a reflective treatment and having a focal point of light is positioned thereon or in a vicinity thereof, and a tip surface that enables incidence of light in a direction different from the reflective control surface, and the shade is made of a transparent material.
